

Regression Error, Skewness, and XGBoost Analysis Report

AI Coverage Data - Revenue Impact Regression

Field	Value
Report date	June 19, 2026
Scope	Error extraction, skewness, percentile ranges, nonlinearity checks, decision tree, and XGBoost interpretation
Source	Exported regression dashboard CSV files in templates/dashboard_downloads

Core reading

Final key takeaway: This is a diagnostic improvement story, not a model-replacement story. XGBoost raises held-out R Square from 0.5170 to 0.5260, while the evidence still points to the same core driver hierarchy: cost savings first, AI investment second, and operational maturity variables as refinements.

Read this as a layered diagnostic: residuals identify where the regression misses, skewness explains why those misses sit in the tail, correlations show the shape of association, and tree models turn that shape into thresholds.

1. Analysis Structure

Stage	Definition	Why it matters
Error feature extraction	Actual, predicted, signed error, absolute error, error percentage, direction	Shows whether failure is random or patterned
Skewness and percentiles	P1-P99, median, skewness, direction	Shows whether averages hide long-tail behavior
Correlation nonlinearity	Pearson vs Spearman and nonlinear fits	Shows whether straight-line correlation is enough
Decision tree	Single skew-aware threshold tree	Turns model shape into readable regimes
XGBoost	Many shallow threshold trees with gamma split control	Tests whether nonlinear rules improve held-out fit

Final key takeaway

Final key takeaway: Read the analysis in order. Error extraction tells us where the model fails, skewness explains why those failures sit in the right tail, correlation checks justify threshold modeling, and the tree/XGBoost layer turns that structure into interpretable split rules.

2. Model Score Context

R Square measures explained held-out variation. RMSE measures average error in revenue-impact units. XGBoost is best by R Square, but the improvement is modest.

Model Comparison: Held-Out Test R Square

XGBoost is best, but its lift over linear regression is modest.



Model	Test R Square	Reading
Linear regression	0.5170	Strong baseline
Single decision tree	0.4728	Readable, weaker fit
XGBoost	0.5260	Best held-out score
XGBoost gain	+0.0089	1.7% relative lift

Final key takeaway

Final key takeaway: XGBoost is the selected model because it has the best held-out score, but the lift is modest: +0.0089 R Square, about 1.7% relative to the linear baseline. Example: RMSE improves by only 26.21K, so the tree layer should be used mainly for nonlinear explanation and targeted refinement.

3. Error Feature Extraction

Signed error is actual minus predicted. Positive error means underprediction. The largest misses are concentrated in high-upside cases.

Largest Error Cases

The largest misses are high-impact underpredictions in the right tail.



Case	Actual	Predicted	Error	Direction
91943	123.42M	14.81M	108.61M	Underpredicted
55570	84.10M	11.32M	72.78M	Underpredicted
48452	93.13M	21.71M	71.42M	Underpredicted
24692	79.74M	9.42M	70.32M	Underpredicted
196367	76.02M	11.37M	64.65M	Underpredicted

Final key takeaway

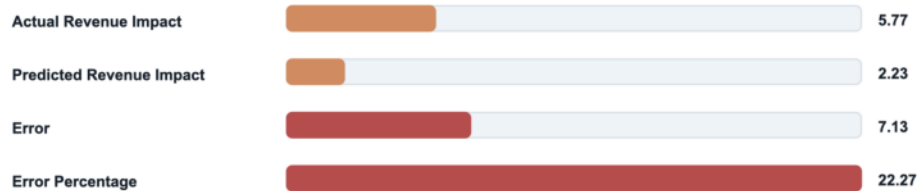
Final key takeaway: The largest errors are not random-looking small misses. They are high-upside underpredictions. Example: case 91943 has actual impact 123.42M, predicted impact 14.81M, and error 108.61M.

4. Skewness and Percentile Ranges

Positive skewness means a long right tail. Percentiles show how far the tail is from the ordinary middle.

Target and Error Skewness

Error percentage and signed error are heavily right-skewed.



Feature Skewness

Cost savings and AI investment are the high-skew predictors.



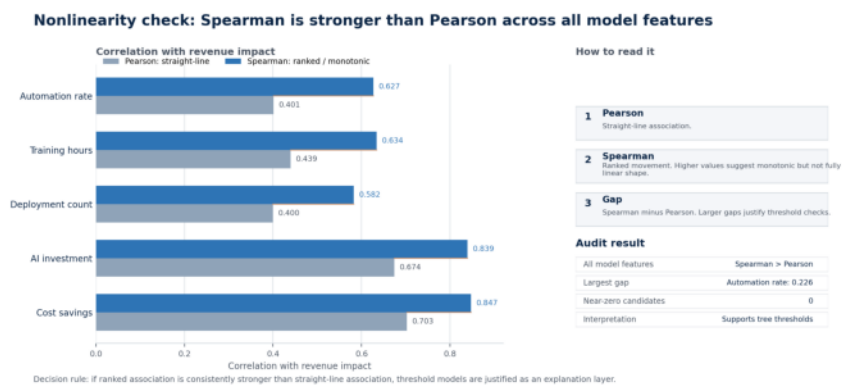
Feature	Median	P99	Skew	Tree input
cost_savings	1.36M	11.88M	3.03	log1p
ai_investment_usd	4.01M	17.77M	1.74	log1p
automation_rate	0.430	0.940	0.16	raw
training_hours	75.5	166.4	0.15	raw
deployment_count	25	51	0.29	raw

Final key takeaway

Final key takeaway: The right tail is the central data issue. Cost savings has skewness 3.03 and P99 11.88M; AI investment has skewness 1.74 and P99 17.77M. These values justify log1p treatment for financial predictors before tree modeling.

5. Correlation Nonlinearity

Pearson checks straight-line association. Spearman checks ranked association. Higher Spearman means ordered but nonlinear shape.



Feature	Pearson	Spearman	Gap	Shape
automation_rate	0.401	0.627	0.226	monotonic nonlinear
employee_ai_training_h ours	0.439	0.634	0.194	monotonic nonlinear
deployment_count	0.400	0.582	0.182	monotonic nonlinear
ai_investment_usd	0.674	0.839	0.165	monotonic nonlinear
cost_savings	0.703	0.847	0.144	monotonic nonlinear

Final key takeaway

Final key takeaway: The current five model features do not hide near-zero Pearson relationships, but they do show monotonic nonlinear structure. The smallest absolute model-feature Pearson is about 0.400; the largest Spearman-Pearson gap is automation_rate at 0.226.

6. Decision Tree Thought Process

A split is a threshold question. A leaf is the predicted value for rows following the full path. The 5% rule stops branches that add too little node-variance gain.

One-Variable Tree Checks

Every variable is tested separately so full feature range remains visible.

cost_savings	ai_investment_usd	employee_ai_training_hours	automation_rate
split: 3.63M gain: 32.9% test R2: 0.318 1.57M -> 7.60M	split: 7.52M gain: 31.0% test R2: 0.299 1.53M -> 7.17M	split: 97.5 gain: 14.1% test R2: 0.139 1.59M -> 4.78M	split: 0.4833 gain: 13.3% test R2: 0.126 1.39M -> 4.31M

Reading: automation_rate and deployment_count matter individually, but are weaker than cost_savings and ai_investment_usd.

They remain in XGBoost as supporting signals even when the pruned interaction tree does not select them.

Node	Rule	Value	Reading
Root	cost_savings <= 3.63M	32.9% gain	Main financial scale split
High branch	cost_savings > 3.63M	7.60M	High-savings rows already high impact
Highest leaf	6,866,957.51 and cost_savings > 12,545,820.55	16.65M	Extreme diagnostic regime

Final key takeaway

Final key takeaway: The single tree is a readable threshold map. The root cost-savings split at 3.63M creates the main scale separation, and the highest diagnostic leaf predicts 16.65M. Treat this as explanation, not the final predictive engine.

7. Full Feature Range

The compact interaction tree shows only the variables that win greedy splits. One-variable trees test whether every feature has standalone threshold value.

Variable	Split	Gain	Test R2	Prediction shift
cost_savings	3.63M	32.9%	0.318	1.57M to 7.60M
ai_investment_usd	7.52M	31.0%	0.299	1.53M to 7.17M
employee_ai_training_h ours	97.550	14.1%	0.139	1.59M to 4.78M
automation_rate	0.483	13.3%	0.126	1.39M to 4.31M
deployment_count	29.500	11.7%	0.112	1.51M to 4.26M

Final key takeaway

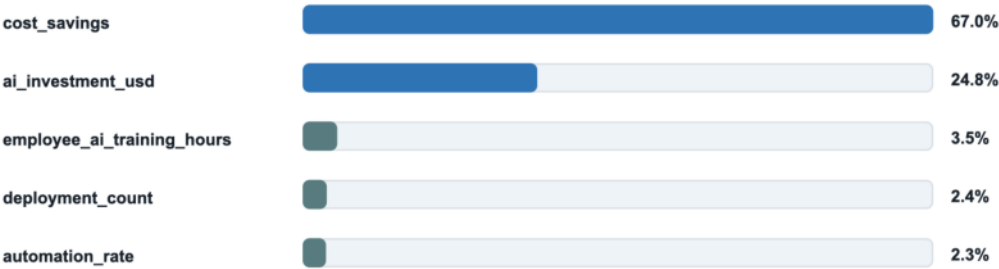
Final key takeaway: Automation and deployment were not disregarded. They simply lost priority inside the greedy interaction tree after stronger financial splits were selected. Individually, automation still splits around 0.430-0.733 scale behavior, and deployment still has a one-variable split at 29.5 deployments.

8. XGBoost Interpretation

XGBoost combines many shallow trees. Each tree adds a correction, and feature importance summarizes contribution to ensemble error reduction.

XGBoost Feature Importance

Financial-scale features dominate; operations provide secondary signal.



Feature	Importance	Input	Skew
cost_savings	67.0%	log1p (high right skew)	2.90
ai_investment_usd	24.8%	log1p (high right skew)	1.68
employee_ai_training_hours	3.5%	raw	0.16
deployment_count	2.4%	raw	0.29
automation_rate	2.3%	raw	0.17

Final key takeaway

Final key takeaway: XGBoost confirms the same hierarchy at ensemble scale. Cost savings contributes 67.0%, AI investment contributes 24.8%, and the three operational variables together contribute 8.2%. The audit shows 1,161 split rows and 0 below the gain threshold.

9. Combined Interpretation

Evidence	Definition	Best use
Coefficients	Controlled linear direction after scaling	Use for baseline comparison
Correlations	Standalone association and rank shape	Use for nonlinear screening
Decision tree	Readable thresholds and leaves	Use for regime explanations
XGBoost	Final nonlinear hierarchy	Use for selected model interpretation

Example: automation rate can have a positive standalone correlation but small controlled/ensemble contribution after investment, savings, deployment, and training are already known.

Final key takeaway

Final key takeaway: Use each evidence type for its proper job. Coefficients explain controlled linear direction, correlations explain standalone association, the single tree explains thresholds, and XGBoost explains final nonlinear hierarchy. Example: automation can have positive standalone association but small controlled/ensemble contribution.

10. Further Analysis and Slide Notes

Next analysis	Question answered	Priority
SHAP values	Which features drive each XGBoost prediction?	High
Percentile-bin analysis	How does impact change across distribution bands?	High
Partial dependence	Are there diminishing returns or threshold jumps?	Medium-high
Interaction checks	Do financial and maturity variables combine?	Medium
Top 1% residual study	What do the largest underpredictions share?	High
Causal design	Do the features cause revenue impact?	Separate project

Final key takeaway

Final key takeaway: The next best analysis is not another broad model comparison. It is case-level explanation of the right tail. Start with SHAP values, percentile-bin plots, and a top-1% residual review to explain why the largest underpredictions remain difficult.

Slide-use takeaway

Final key takeaway: Slides should stay visual and selective. Use one message per slide: model lift is small, errors are right-tail underpredictions, correlations are monotonic nonlinear, the decision tree gives thresholds, and XGBoost confirms the feature hierarchy.

Implementation takeaway

Final key takeaway: The requested framework is implemented as a diagnostic pipeline: skewness and percentile extraction, skew-aware tree inputs, XGBoost with gamma split enforcement, and nonlinear correlation checks. The remaining gap is deeper case-level explanation, not basic implementation.